

Adaptive kNN Using Expected Accuracy for Classification of Geo-Spatial Data

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1 Problem & Motivation

Classification is one of the most important tasks in machine learning. One of the most popular classification methods is the k-Nearest Neighbor (kNN) algorithm, which predicts the class of a point based on its nearest neighbors.

However, standard kNN uses one fixed value of k for all data points. Real-world datasets often contain regions with different densities, so one global value of k may not work well everywhere. Small values of k are sensitive to noise, while large values may oversmooth decision boundaries and reduce accuracy.

This issue is related to the bias-variance tradeoff:

- Small $k \rightarrow$ low bias, high variance
- Large $k \rightarrow$ high bias, low variance

To solve this limitation, the paper proposes ACKER (Adaptive Classifier based on kNN Expected accuracy). Instead of using one fixed value of k , ACKER dynamically selects the most suitable value of k for every point.

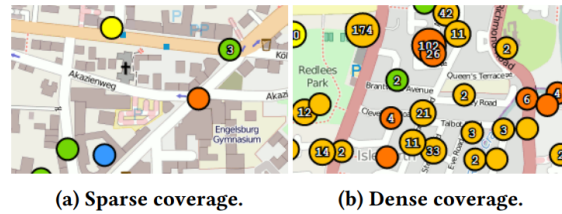


Figure 1: Sparse and dense coverage regions from the paper.

2 Key Idea / Method

The main idea of ACKER is to automatically choose the best value of k separately for each new point. The algorithm evaluates several candidate values of k and selects the one with the highest expected accuracy.

The optimization objective is:

$$\max(\text{exp_acc}(p, k))$$

where:

- p is the point being classified,
- k is the candidate number of neighbors,
- exp_acc is the expected accuracy estimate.

The ACKER workflow consists of the following steps:

1. Receive a new point
2. Try multiple values of k
3. Compute expected accuracy
4. Select the best k
5. Apply standard kNN

Expected accuracy is based on similar regions from the training dataset. The intuition is that if a certain value of k worked well for structurally similar points, it will likely work well for the current point. Expected accuracy is defined as:

$$exp_acc_l(p, k) = acc(\hat{c}_k, C_{sim,l}(p, k))$$

The paper introduces several similarity functions based on neighbor distances, including average distance, maximum distance, and a combined similarity function. The combined approach often achieved the best performance during experiments.

This paper is strongly connected to SDS 302 topics such as nearest neighbor classification, hyperparameter optimization, validation, and bias-variance tradeoff.

3 Results

The authors evaluated ACKER using geo-spatial datasets including Twitter Milan, WideNoise, and San Francisco Crimes datasets. The experiments used 10-fold cross-validation and compared standard kNN with the proposed adaptive method.

The results showed that ACKER generally outperformed standard kNN. One of the strongest improvements was achieved on the Twitter dataset:

Method	Accuracy
Standard kNN	77.3%
ACKER	82.5%

Table 1: Performance comparison between standard kNN and ACKER.

The experiments also demonstrated that:

- adaptive selection of k improves classification accuracy,
- similarity functions influence performance,
- the combined similarity function often produced the best results.

The paper also showed that reducing the range of candidate values of k can reduce runtime without decreasing accuracy.

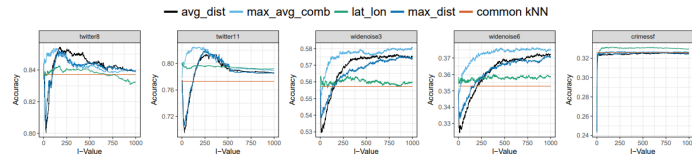


Figure 2: Accuracy of ACKER classifier depending on parameter l .

4 Critical Insight

Strength

One major strength of ACKER is flexibility. Unlike traditional kNN, ACKER adapts to local structures in the dataset by dynamically selecting the most suitable value of k for each point. This improves classification accuracy in datasets with irregular densities.

Limitation

The main limitation of ACKER is computational complexity. The algorithm must test multiple values of k , compute expected accuracy, and search for similar points, making it slower than standard kNN.

Another limitation is that the experiments mainly focused on geo-spatial datasets. More testing on high-dimensional datasets is necessary.

5 Conclusion

This paper presents an adaptive improvement to the traditional kNN classifier. Instead of using one fixed neighborhood size, ACKER dynamically selects the most suitable value of k using expected accuracy estimation. Experimental results showed that this adaptive approach improves classification accuracy while maintaining the simplicity and interpretability of kNN.

References

- [1] M. Kibanov et al., *Adaptive kNN Using Expected Accuracy for Classification of Geo-Spatial Data*, SAC 2018.